

FORMS AND FUNCTIONS IN PLANT

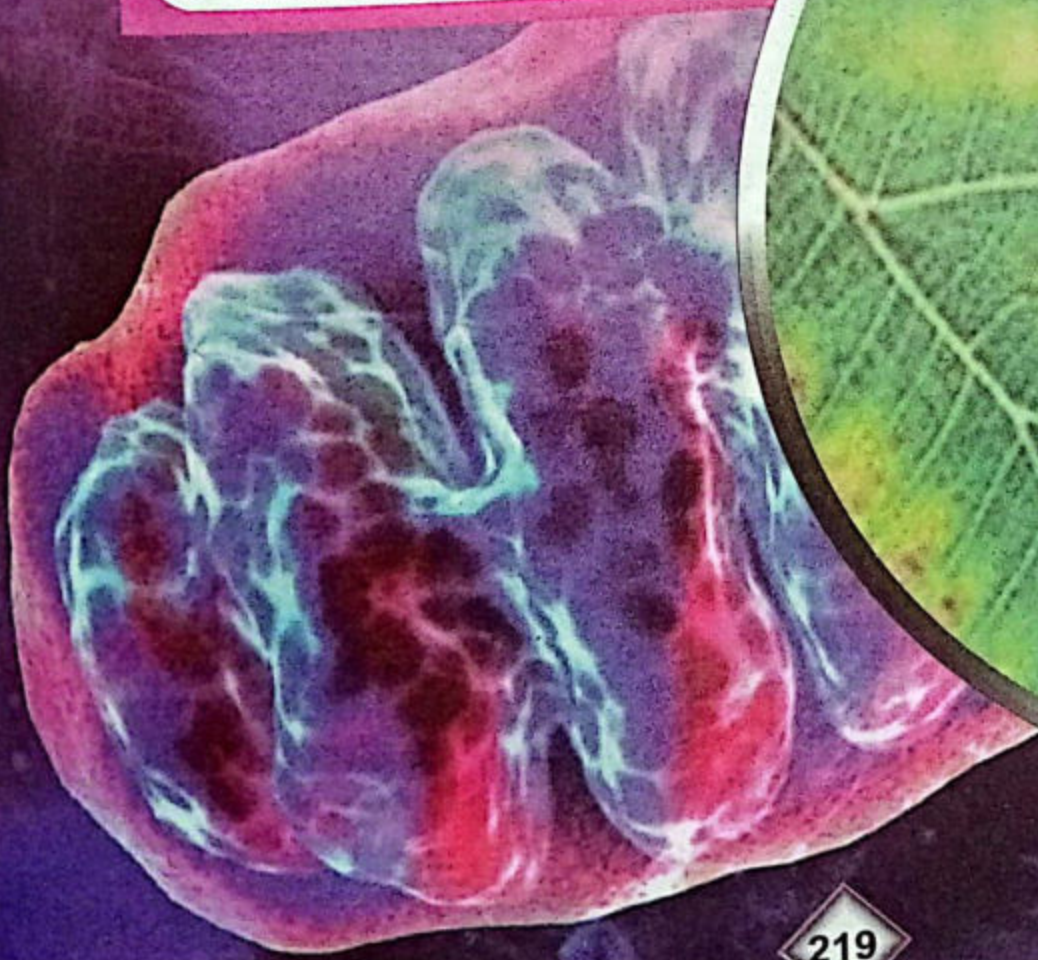
Chapter

10

Major Concept

In this Unit you will learn:

- Nutrition in Plants
- Gaseous Exchange in Plants
- Transport in Plants
- Homeostasis in Plants
- Support in Plants
- Growth and Development in Plants
- Growth Responses in Plants





Introduction:

All living organisms must obtain a variety of chemical substances from their surroundings. These substances are essential in order to build up their body. Living organisms also required energy to perform their functions. Nutrition is the process of acquiring energy and materials.

Plants obtain essential nutrients primarily from soil. Nutrient cycle comprises from plants to decomposers and decomposers to soil thus back of plants. Nitrogen in the atmosphere becomes available to plants as nitrates and ammonium by nitrogen fixing bacteria.

Plants are photosynthetic autotrophs. They prepare their food by photosynthesis. In addition to light, carbon dioxide and water, which are required for photosynthesis, plants use other raw materials to synthesize the organic compounds as a secondary products for growth. These elements are known as mineral nutrients. Plants use minerals like nitrates, sulphates and phosphates to make other secondary products like proteins, lipids that are needed for growth and other metabolic activities.

These nutrients are inorganic elements, 14 of which are essential, especially nitrogen (N) phosphorus (P), potassium (K) and magnesium (Mg) etc.

All plants appear to have very similar nutritional requirements, although the amounts required, vary between individual plants and between species. In general, macronutrients are required in large amounts, Nitrogen (N), Phosphorus (P), Potassium (K), Sulphur (S), Sodium (Na), Chlorine (Cl), Magnesium (Mg), and Calcium (Ca) are included in macronutrient, Micronutrient (trace elements) are required in small amounts including Manganese (Mn), iron (Fe), Cobalt (Co), Copper (Cu), Zinc (Zn), Molybdenum (Mo), Boron (B), Fluorine (F) and Iodine (I).

10.1.1 Role of some important minerals nutrients

Analysis of plants shows the presence of a large number of mineral elements. The amount and number of elements presence in plant may differ from plant to plant, place to place and medium to medium in which plant grows. The role of some important macronutrient and micro nutrients in plants discuss below.



Role of macronutrients and mineral nutrients in plants and their deficiency symptoms:

Important nutrients	Major Role	Deficiency symptoms
Nitrogen	Essential constituent of proteins, nucleotides, nucleic acids, chlorophyll etc.	Leaves turn pale yellow due to loss in chlorophyll content and cell enlargement are inhibited. Plants growth remain stunted.
Phosphorous	Promotes healthy root growth and fruit ripening, involved in the formation of cell-membrane as phospholipids, nucleic acid, co-enzyme (NAD and NADP) ATP role in energy transfer reaction.	Growth is retarded. Necrosis dead cells patches appear on leaves, petioles and fruits. Cambial activity is checked.
Potassium	Role in the stomatal opening and closing, enzyme activator in synthesis of certain peptide bonds and carbohydrate metabolism.	Leaf turns into dull or bluish green, chlorosis, growth is stunted, shortening of internodes.
Sulphur	Essential for growth and development of all crops, used in the formation of amino acid, proteins and oils, helps to develop and activate certain enzymes and vitamins.	Plants are small and spindly with short and slender stalks; their growth is retarded and nitrogen - fixation reduced.
Magnesium	Activates enzymes, used in the formation of chlorophyll and some carbohydrates, conversion of starches to sugar and its presence in plant tissue helps the plant to withstand in cold temperature.	yellowing of leaves, often interveinal, death of leaf tissue on areas of chlorosis, chlorotic areas may turn brown colored.

10.1.2 Some micronutrients and their role:

i. **Manganese (Mn):**

Manganese are absorbed by plants in very minute quantity and become the part of enzyme phosphotase (transfer PO_4 groups). Manganese involved in oxidation reduction reaction.

ii. **Iron (Fe):**

Iron can exist in the forms Fe^{2+} and Fe^{3+} is a constituent of cytochromes, which are involved in the electron transport reactions of photosynthesis and respiration.

iii. **Zinc (Zn):**

The role of zinc in anaerobic respiration in plants (alcohol fermentation)

Essential Elements	
Non metallic	Metallic
Carbon	Potassium
Hydrogen	Calcium
Oxygen	Magnesium
Nitrogen	Iron
Phosphorous	
Sulphur	

10.2 CARNIVOROUS PLANTS (INSECTIVOROUS PLANTS)

Insectivorous or carnivorous plants are green plants which are specially adapted for trapping and digesting small animals, particularly insects. Such plants typically live in nitrogen deficient habitat, and use the animals principally as a source of nitrogen. They trap the insects with their colours, scent or sweet secretion. Then secrete enzymes and carry out extra cellular digestion of nitrogenous compounds. The products notably amino acids are absorbed and assimilated. Some common examples are nepthens, pitcher plant, *Drosera intermedia* or sundew, *Dionaea muscipula* or Venus fly trap and *Aldrovanda* (water fly trap)

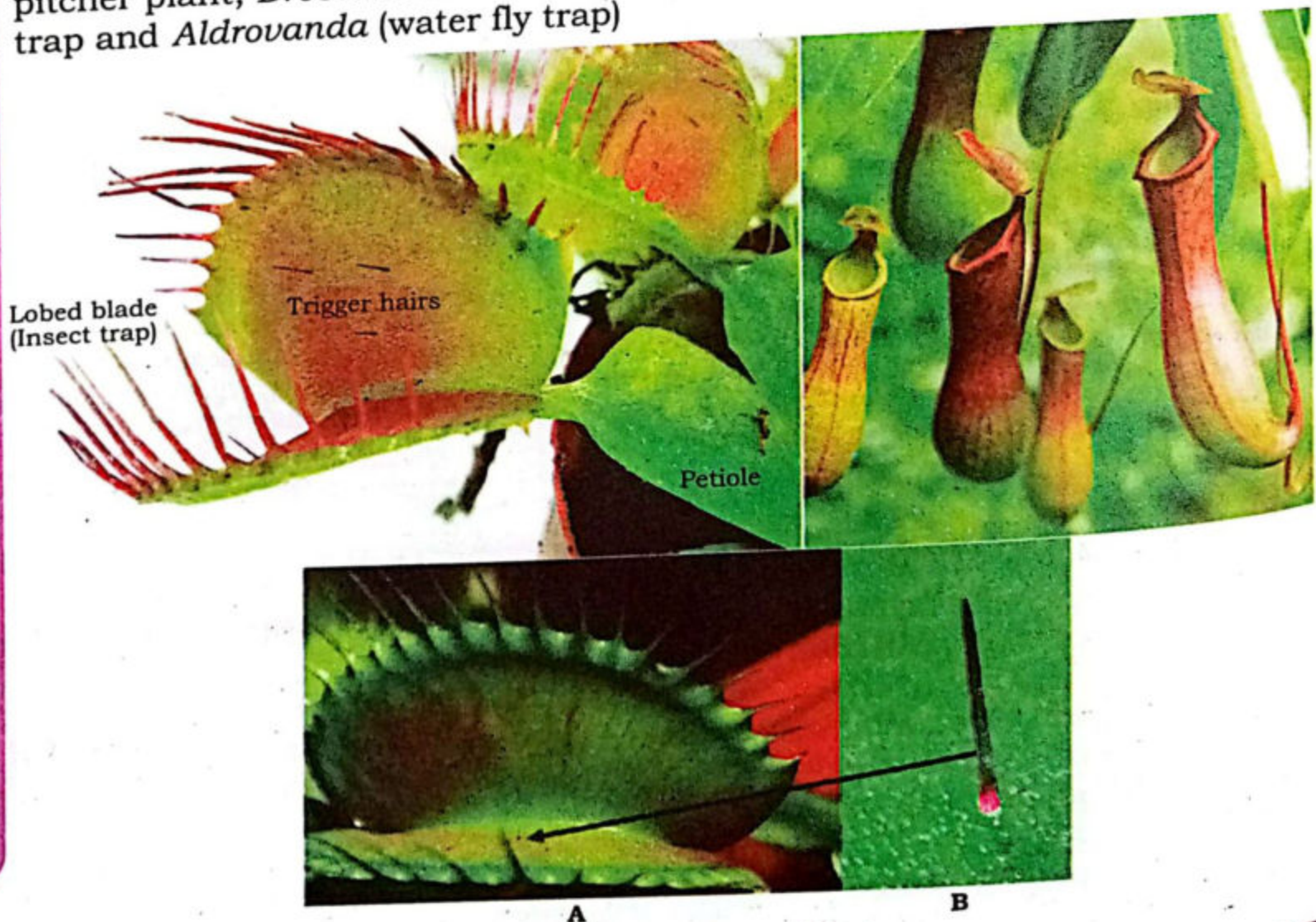


Fig: 10.1. Carnivores Plants



10.3 Gaseous Exchange in Plants:

All plants exchange gases for respiration as well as for photosynthesis. The process of respiration occurs constantly day and night in all living cells of plants. Oxygen is being continually absorbed whereas carbon dioxide is liberated. During photosynthesis intake of carbon dioxide take place through the stomata and lenticels, while oxygen is liberated during this process photosynthesis occur during day time and in chlorophyll containing cells only.

Gaseous exchange in unicellular organism and lower plants especially in the aquatic ones, takes place by diffusion across a moist cell membrane and wet body surface respectively. Surface area in such organisms is greater than their volume so diffusion alone is sufficient for the transport of gases from surface cells to inner ones. But in higher land plants gas exchange occur by the entire surface of cells present inside the leaves.

Flowering plants exchange gases by diffusion through pores called stomata in their leaves and green stem. Each stoma is formed by two modified epidermal bean-shaped guard cells. Unlike other epidermal cells guard cells bear chloroplast with thicker inner and thinner outer walls. Stomata can be opened or closed depending upon the turgidity of guard cells which is regulated by concentration of sugar and potassium K^+ ion.

Stomata regulate the passage of carbon dioxide, oxygen and water as well as concentration of sugar and potassium across the surface of the leaf.

Leaves are the main sites of gaseous exchange. There are two types of leaf **monofacial** and **bifacial**. In bifacial leaves the upper region of leaf consists of an upper epidermis covered with thick cuticle and epidermis contains less number of stomata. While lower region of leaf consist of lower epidermis with abundant stomata and spongy mesophyll cells, which functions in gaseous exchange and the regulation of transpiration. Inside the leaf of a dicotyledon(bifacial) there is a spongy mesophyll with large air spaces, which connect with each other and those between the palisade cells. This system of air space allows gases to diffuse freely between the cells. The mesophyll tissue of the leaf possesses a continuous system of intercellular spaces which connects with the external atmosphere through the stomata, which is suited for the diffusion of air into the intercellular spaces.

The intensity of gaseous exchange depends upon the intensity of respiration. Though all living cells of different organs like stem root etc. constantly respire, yet there is no transport system within plants for respiratory gases. Such gases are therefore transported by diffusion only, in order to facilitate the process of diffusion of gases, plant tissues are permeated by air spaces.

Once the air gets inside the plant, movement of oxygen is determined by the diffusion gradients that exist in the intercellular air spaces. In this



way oxygen travels towards the cell and dissolves moisture present on the surface of cell. From here oxygen passes by diffusion into the cells themselves. Carbon dioxide leaves the plant by the same pathway but in the reverse direction. The whole situation becomes more complex in photosynthesizing parts. Here oxygen is also produced by the chloroplast as a waste product of photosynthesis. The oxygen may be used up immediately in respiration by mitochondria in the same cell, and waste carbon dioxide from respiration may be used by the Chloroplast, for photosynthesis.

Plants perform photosynthesis and respiration both during the day, but in night only respiration does occur, so methodology of gaseous exchange is different in day and night.

During day the rate of photosynthesis is much higher than the respiration, concentration of carbon dioxide release during respiration is very low which does not meet the requirements of photosynthesis. Plants obtain required carbon dioxide for photosynthesis from environment; In the same way the oxygen released during photosynthesis is much more than the requirements of respiration so plants released it from its body.

At night plants absorb oxygen and release carbon dioxide. Thus, in night the carbon dioxide is in excess and oxygen is in deficient.

The plants which growing in well watered conditions a diurnal stomatal rhythm is observed, stomata are open in day closed in night. The factors which influence the opening and closing of stomata are light, concentration of Potassium (K^+) ions and concentration of carbon dioxide in the air. Stomata are sensitive to CO_2 concentration in the surrounding air.

10.3.1 Transport in Plants:

Every living cell, whether it exists alone as a single cell organism or as a component of a multicellular one, must perform its own metabolic activities. It must synthesize its own ATP by cellular respiration. Every cell must obtain the necessary raw materials to support its metabolism for the synthesis of ATP. Obviously, some mechanism is needed for transporting substances between the specialized system of procurement, synthesis or elimination of different compounds.

Plants are in contact with both soil and atmosphere. Various materials from atmosphere and soil are transported in and out of plant body by various process such as diffusion, osmosis, imbibition and active transport.

In plants, cell to cell movement of water and solutes occur via apoplast and or symplast pathways. Long distance transport occurs in xylem and phloem. Transport mechanism include passive processes such as diffusion, mass flow and osmosis and active processes, which require energy from cell metabolism.



Non-gaseous substances that are transported into and within plants are water and solutes such as ions, amino acids and sucrose. There are number of pathways:

- Between solutions in the external environment and plant roots
- Between cells either along the apoplastic pathway or the symplastic pathway via plasmodesmata
- Between compartments within a cell, for example between vacuole and cytoplasm which are separated by a membrane, the tonoplast.
- Long distance transport in xylem and phloem.

Movement of water and solutes takes place through Xylem and mass flow transport of sugar through Phloem. Both Xylem and Phloem are structurally related with their functions.

Xylem:

Xylem is a complex tissue consisting of four types of cells in angiosperm out of them two are water conducting cells. These are open ended cells called vessels and porous cells called **trachieds**.

Vessels:

Vessels are thick wall tube like structures which extend through several feet of Xylem tissues. They range in diameter from 0.2 mm to as much as 0.7 mm. Their walls are lignified and perforated by pits. At the pit lignin is not deposited and cell wall is thin made up of cellulose. The pits match up with pits of adjacent cells so that cell cavities are connected to adjacent cells cavities. Xylem vessels arrive from cylindrical cells, which are placed end to end. At maturity, the end walls of these cells are dissolved and cytoplasmic contents die. Thus a continuous duct is formed which offers a better route for long distance transport of water from roots upto leaves. The rate of flow of water in vessels is 10 times faster than trachieds. Vessels are mostly found in flowering plants.

Trachieds:

Trachieds are unusual cells about 0.3 mm in diameter and several mm in length. They can be distinguished from vessels by their angular walls and smaller size. They taper at each end and taper ends of one cell overlap taper ends of other cells, like xylem vessels they are dead with thick lignified walls.

Their walls are perforated by small holes called pits which are of two types; simple and bordered. The pits in the cross walls connect upper trachieds with lower one. Through these pits water and minerals flow freely from one trachied to another. In all tracheophytes except angiosperm trachieds are the only water conducting ducts.

Phloem:

Phloem consists of four types of cells

- | | | | |
|------|---------------------|-----|-------------------|
| i. | Sieve tube elements | ii. | Companion cells |
| iii. | Phloem fiber | iv. | Phloem parenchyma |

Sieve tube elements:

Sieve tubes are the long tube like structures that transport photosynthetic products throughout the plants. They are formed by the end to end fusion of cells called sieve tube elements. Sieve tube elements have a very distinctive structure. Their walls are made of cellulose and pectic substances, their nuclei degenerate and are lost as they mature and the cytoplasm becomes confined to a thin layer around the periphery of the cell.

They have thin primary cell wall that connect adjacent cells by their ends. They have perforation in their transvers wall, which look like the holes in a sieve. Sieve cells are elongated and form a continuous conducting pathway throughout a plant.

A conspicuous and characteristics feature of sieve tubes is the sieve plate. This is derived from the two adjoining end walls of neighboring sieve elements.

Companion cells:

In flowering plants, each sieve cell is associated with a specialized parenchyma cell a companion. The two cells together form a functional unit. Companion cells retain all their cytoplasmic contents and have a well-developed endoplasmic reticulum with a large dense nucleus. The function of the companion cell is to control the activity of the sieve cell.

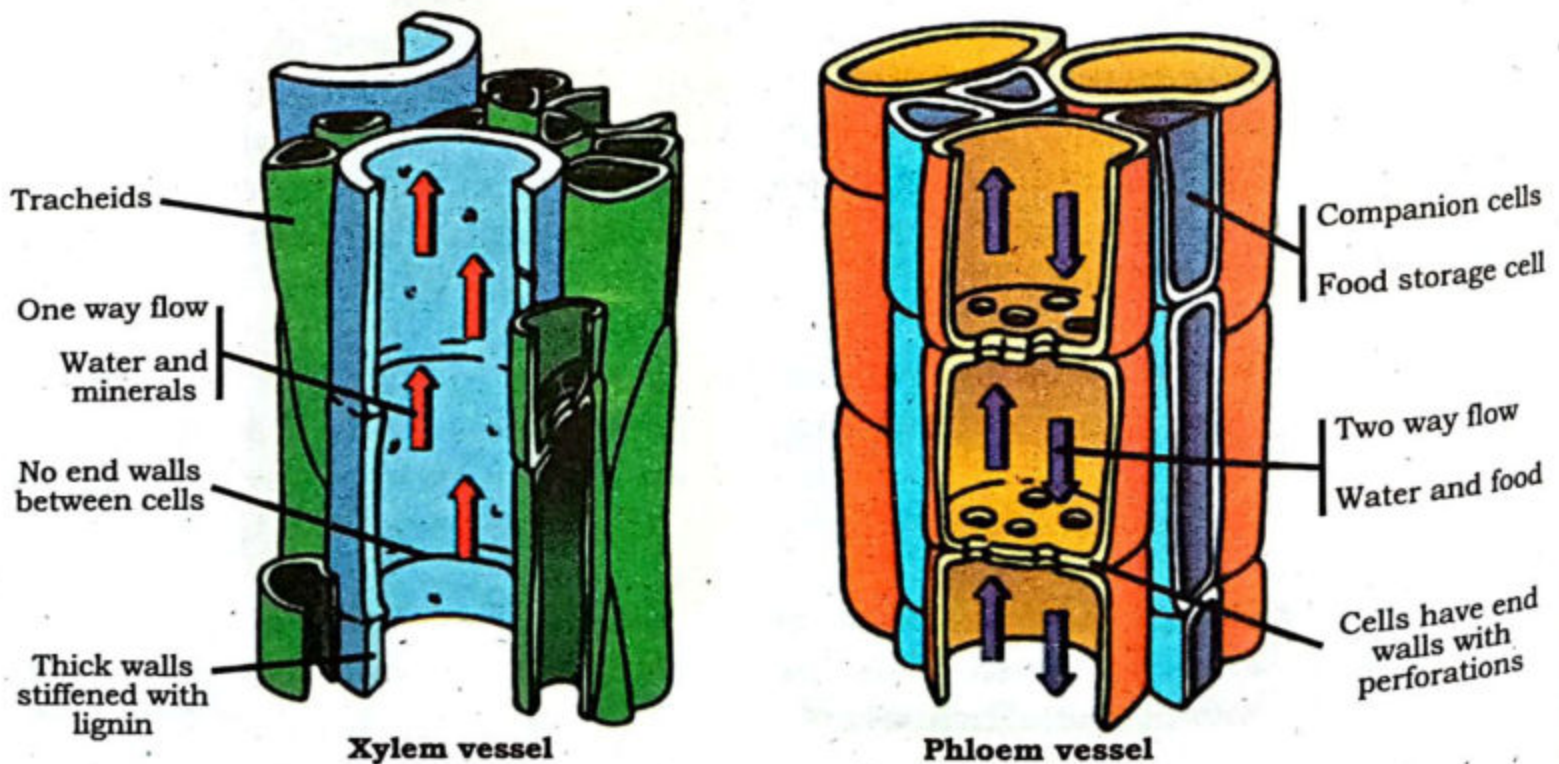


Fig: 10.2. Xylem and phloem



10.3.2 Water transport:

The movement of water molecules from soil into roots, or from leaf mesophyll cells through stomata to the environment, takes place regularly. Water moves from soil into roots and up to aerial parts of plant by diffusion, osmosis, bulk flow or some combination of their fundamental transport mechanism. The movement of water in or out of the cell exhibit three types of water relations i.e water potential, Osmotic pressure and solute potential.

i. **Water potential:**

The chemical potential of water is a quantitative expression of the free energy associated with the water. Thermodynamically, free energy represents a potential for performing work. All living things including plants, require a continuous input of free energy. In the case of water movements this free energy is involved in water flow. The unit of chemical potential is energy per mole of a substance (joules per mole).

For practical reason, it turns out that the unit of chemical potential is inconvenient for most work in plant physiology. Therefore, plant physiologists have defined another parameter called **water potential** as the difference between the free energy of water molecules in pure water and energy of water in any other system (e.g. water in solution or in cell sap of plant). Now, the free energy of water is expressed in pressure unit such as megapascals and symbolized by Greek letter ψ (MPa: MPa=9.87) atmosphere).

Pure water has been assigned the value of water potential "0" MP. Addition of solute particles lowers the mole fraction (number of mole of substance divided by total number of all substances in the system / solution of water. Hence, there is a decrease in water potential. Therefore, values of water potential remains less than zero or in negative value.

ii. **Osmotic pressure:**

The pressure exerted upon a solution to keep it in equilibrium with pure water when the two are separated by a semipermeable membrane is known as **osmotic pressure**. Therefore, the osmotic pressure of a solution is a measure of the tendency of water to move by osmosis into it. In other words we can say that the osmotic pressure is the pressure that must be exerted on a solution to prevent the passage of solvent molecule into when the solvent and solution are separated by a differentially permeable membrane. Thus, it prevents the process of osmosis proceeding.

iii. **Osmotic Potential or Solute Potential:**

Osmotic potential is the tendency of a solution to attract water molecules when the solutions of two different concentrations are separated by a differentially permeable membrane. Pure water is assigned the osmotic potential zero as the highest value. Since the osmotic potential decreases as the osmotic concentration (the no: of osmotically active particles per unit volume) increases, all solutions have value of less than zero. Under constant

temperature and pressure, water moves from the solution of lower osmotic potential to the solution of higher osmotic potential when the two solutions are separated by a differentially permeable membrane. It is represented by ϕ or solute potential.

Another term used in relation to water potential is pressure potential, which is defined as the hydrostatic pressure in excess of atmospheric pressure.

Water relations of Plant Cell:

For practical purposes a plant cell can be divided into three parts: (i) **Cell Wall:** This is non-living, permeable, outer most boundary of cell made up of cellulose. (ii) **Cytoplasm** along with nucleus forms protoplasm-the lining material bounded by cell membrane. (iii) In the centre, there is a vacuole enclosed by tonoplast; central vacuole is filled with cell sap-an aqueous solution of salts, organic acids and sugar.

The presence of solute particles lowers the water potential ϕ of cell sap. Greater the number of solute particles, the more negative will be the water potential of cell sap. The concentration of solute particles in a solution is known as **solute potential** ϕ_s (This has been previously referred as osmotic potential). The value of solute potential is always negative.

When a cell is placed in pure water or in an aqueous solution with higher water potential (less negative) than the cell sap, water flows into the vacuole by osmosis through plasma membrane and tonoplast. As more water flows into the vacuole, the tension developed by cell wall causes an internal hydrostatic pressure to develop. This is called **pressure potential** ϕ_p and it opposes the continued uptake of water into the cell by osmosis. When the cell wall is fully stretched and pressure potential reaches at its maximum, the cell cannot take any more water, and is said to be fully **turgid**. The relationship between water potential ϕ_w , solute potential ϕ_s and pressure potential ϕ_p is represented by following equation.

$$\phi = \phi_s + \phi_p$$

In a turgid cell ϕ_p is equal and opposite to ϕ_s so $\phi = 0$

10.4.1 Water and minerals uptake by root:

Absorption of water and minerals salts takes place through root system. Roots are provided by enormous number of tiny root hairs which are outgrowths of epidermal cells and found at the root tip. The root hairs greatly increase the surface area of root. Because of large root surface area plants absorb enough quantities of water and inorganic ions for their survival and growth.



and water moves into them by osmosis. From pericycle water flows into Xylem both via symplast and apoplast.

Diffusion moves water from region of high water concentration (low solute concentration) to region of low water concentration (high solute concentration). It occurs because molecules are in constant thermal agitation. Bulk flow occurs in response to a pressure difference. Whenever, there is a suitable pathway for bulk movement of water. Osmosis is the process by which water moves across differentially permeable membrane. It depends upon the chemical potential of water or water potential. Osmotic potential of two separating solution and pressure potential across the membrane and wall of the cells. Therefore, it is necessary to understand these terms to explain the process of water movement across the membrane.

10.4.2 Movement of water in Xylem:

Plants absorb large quantity of water from soil. Only 1-2% of the absorbed water used in photosynthesis, in other metabolic activities and in the maintenance of turgidity of the cells. The remainder is lost from the leaves and other aerial parts in the form of vapours. This loss of water in vapour form through aerial parts of plants body is called **transpiration**. Before transpiration water is conducted up to leaves through xylem. This upward movement of water from absorptive surface (roots) up to transpiring surface(leaves) against the downward pull of gravity is known as **ascent of sap**.

The upward movement of water from roots upto leaves takes place through xylem depend upon two factors. These are transpiration pull and physical properties of water i.e., adhesion and cohesion. Water molecules tend to adhere to cellulose molecules of the walls of xylem vessels. Extensive hydrogen bonding in water gives rise to property of cohesion. Its also gives water a high tensile strength defined as the ability to resist the pulling force. The water molecules in xylem vessels forms a continuous column.

Transpiration pull results chain of events that starts when leaves begin to absorb solar radiation in the morning. Sunlight raises temperature of leaves, so the water begins to evaporate from moist walls of mesophyll cells. The evaporated water is immediately replaced from water inside the cell which is replaced with water from neighboring cell deeper in the leaf. Ultimately, water is pulled from xylem to meet the loss of water. Thus, water in xylem is placed under tension which is transmitted to root through vessels. This downward transmission of tension is because of cohesive property of water columns in vessels and trachieds. Water column moves upward by mass flow due to transpiration pull. Adhesion of water molecules to hydrophilic walls of Xylem cells. Small diameter of vessels and trachieds are important factors in overcoming the force of gravity.



To transport water over a long distance, plants do not use their metabolic energy. Force like adhesion, cohesion and evaporating effect of sunlight are mainly responsible for upward conduction of water.

Mechanism of opening and closing of stomata:

The opening and closing of stomata and even widening and narrowing of the gap between two guard cells depend upon the turgidity of guard cells, which is due to increase or decrease in the osmotic potential of the guard cells.

The change in water potential that result from the osmotic changes cause water to move in or out of the guard cells. If water moves in, the cells expand (become turgid), if water moved out, cells become flaccid. When guard cells are turgid stomata open. When guard cells are flaccid the stomata are closed to affect this movement of water and exchange must take place between the guard cells and surrounding mesophyll and epidermal cells.

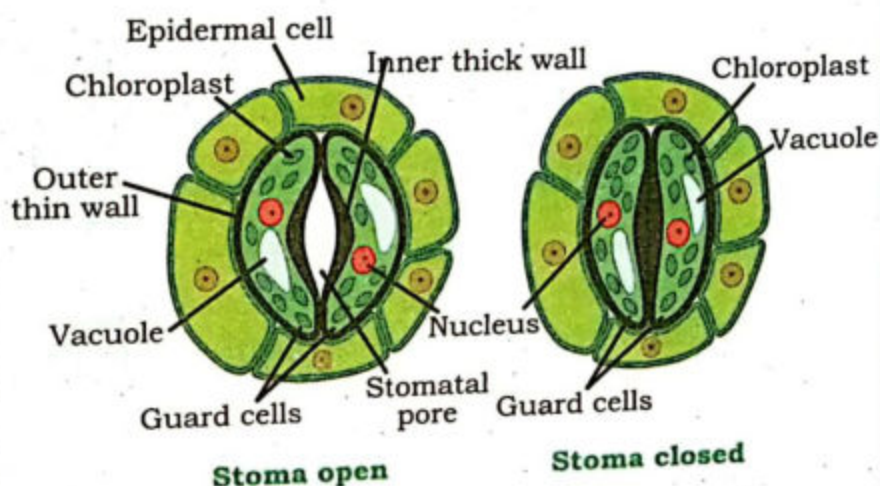


Fig: 10.4. Stomata

10.5.1 Translocation of organic solutes (phloem translocation):

The product of photosynthesis move from mature leaves to the growing and storage organs of plants. The direction of transport is determined by the relative locations of the sources and sinks of photosynthates. This movement of photoassimilates and other organic materials take place via the Phloem, and is therefore called **Phloem Translocation**. Transport occurs through specialized tissues called Sieve elements.

Source-sink movement:

The translocation of photosynthates always takes place from source to sink tissues, therefore, this Phloem transport is also referred as **source-to-sink movement**. Source are those tissues which produce photoassimilate more than their needs and sink are those tissues which produce photoassimilates less than they need or do not produce photoassimilates, like the mesophyll cells all of middle parts of leaves are source and the fruits, seeds or roots are sink. This pathway follows developmental changes as some sink and source tissues are inter convertible during the development of the plant e.g. developing and germinating seeds, developing and mature leaves.

A number of steps involved in the movement of photosynthates from mesophyll chloroplast to the Sieve elements in the Phloem of mature leaves. Sucrose is synthesized in the cytosol of mesophyll cells. This sucrose is translocated from the mesophyll cells to the vicinity of the sieve elements in the smallest veins of the leaf. This generally termed as the short distance transport pathway because the solutes cover only a distance of two or three cell diameters. The sucrose is then actively transport into Sieve elements in the steps commonly called **Phloem loading**.

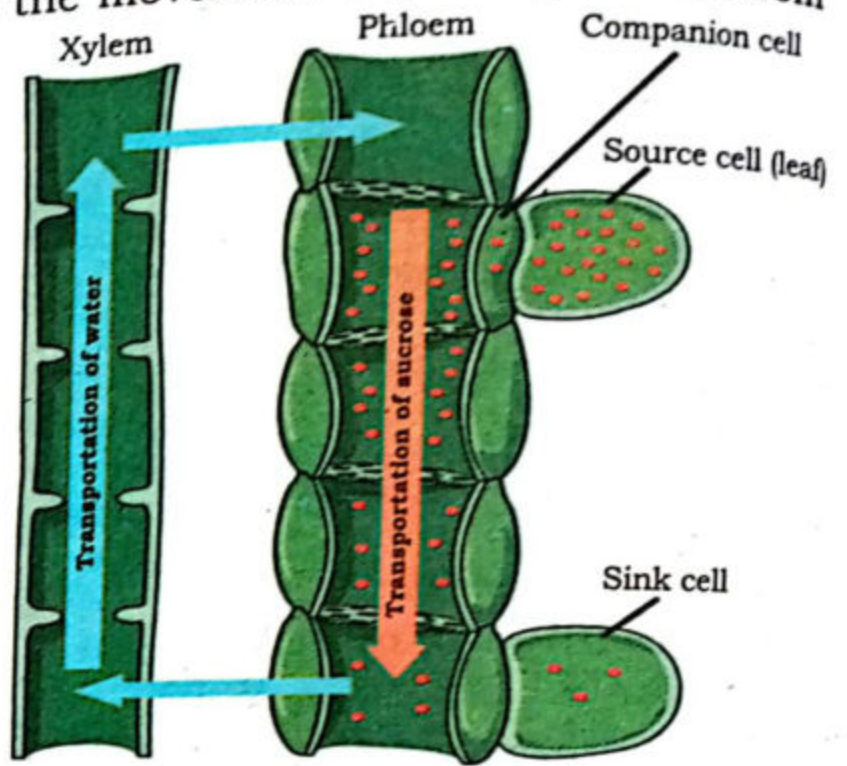


Fig: 10.5. Phloem translocation

The pathway of Phloem loading may be either symplastic or apoplastic depending upon the species.

The sucrose in sieve elements is exported away from source tissues. The photoassimilates can be moved a long distance hence this is termed as long distance transport. Finally, the photo assimilates or sucrose is unloaded at the sink in a process referred to as **Phloem unloading**.

10.5.2 The mechanism of phloem translocation:

Pressure flow hypothesis:

Phloem translocation is mainly explained by a theory called the pressure flow hypothesis proposed by **Ernst Münch** in 1930, which states that the flow of solution in the sieve elements is driven by a pressure gradient produced due to difference in osmotic pressure between sources and sinks. This pressure gradient is produced due to Phloem loading and unloading at the source and sink, respectively. A high osmotic pressure generated due to active Phloem loading in the sieve elements of a source tissue, cause a decline in the water potential. Whereas the xylem has high water potential so water moves out of xylem and enter into phloem and produces a high turgor pressure. In the sink tissue, present at other terminal of the translocation pathway, phloem unloading occurs, which produces a low osmotic pressure in the sieve elements of the tissues. As the low osmotic pressure, the water potential of the phloem rises above that of the xylem and water tends to leave the Phloem in response to the water potential gradient, which causes the decrease in the turgor pressure of the Phloem sieve elements in the sink.

An equilibrium between the two ends (source and sink) would be reached very soon if the entire translocation pathway were a single



membrane bound compartment. The sieve plates which are present in the sieve elements increase the resistance along the pathway and maintain the pressure gradient in the sieve elements between the sources and sinks. The photoassimilates present in sieve elements are physically pushed along the translocation pathway by bulk flow, much like water flowing through a garden hose.

Water movement in the translocation pathway is therefore driven by the pressure gradient rather than the water potential gradient. The passive, pressure driven long distance translocation in the sieve tube ultimately depends on the active short-distance transport mechanism involved in the phloem loading and unloading. These active mechanisms are responsible for setting up the pressure gradient in the first place.

10.6 HOMEOSTASIS IN PLANTS:

10.6.1 Osmotic adjustment or osmoregulation:

Changing in environmental conditions are the big threat for plants and cause severe stresses on them. Plants cope these stresses by adaptation or adjustment that enable them to survive in the stressful condition. They require metabolism to maintain internal condition of plant. This set of metabolic function is called homeostasis. Osmotic adjustment is a beneficial mechanism for water stress and salinity. It is a useful drought tolerance in response to drought condition or saline stress, many plants accumulate organic solutes to lowering the osmotic potential. Osmoregulation is one of the important aspects of **homeostasis**. It involves maintaining a balance between water and solute contents of cells.

In case of plant cells, the water potential of cell sap (solution in vacuole) is termed as **solute potential**. If a plant cell is placed in pure water or solution of higher water potential than the solution in its vacuole, the water moves from outside to inside the cell (endosmosis) and ultimately into the vacuole. As a result, the cells swells or become turgid. Such an extreme solution is called **hypotonic**. Further allowing the cell into such hypotonic medium doesn't cause it to burst because a cell's wall develops a tension causing an internal hydrostatic pressure or pressure potential due to which further uptake of water in cytoplasm is resisted and finally stops.

When plant cell is placed in concentrated solution or **hypertonic** solution, there is a net movement of water out of the cell. As a result, the cell becomes flaccid. Under such condition, the cytoplasm with its plasma membrane shrinks from the cell wall. This condition is called **plasmolysis**.

Osmotic adjustment:

Depending upon the availability of water to flowering plants in their natural habitat they are grouped into four categories;

- | | |
|-----------------|----------------|
| i. Hydrophytes | ii. Halophytes |
| iii. Mesophytes | iv. Xerophytes |

Hydrophytes:

The plants which are found in fresh water habitat either partly or completely submerged are termed as **hydrophytes**. They do not have any difficulty in obtaining water. The stems and leaves of hydrophytes generally lack cuticle in partially submerged hydrophytes. Stem is cylindrical and spongy, stomata are restricted on the upper surface of leaves. E.g., Water Lilly.



Hydrilla



Vallisneria



Utricularia



Chara



Ceratophyllum

Fig: 10.6. Submerged Hydrophyte

In completely submerged hydrophytes (such as hyderilla) roots are poorly developed. Stem is soft, spongy weak and cylindrical. Internally the cortex contains large air cavities, these tissues are called aerenchyma.

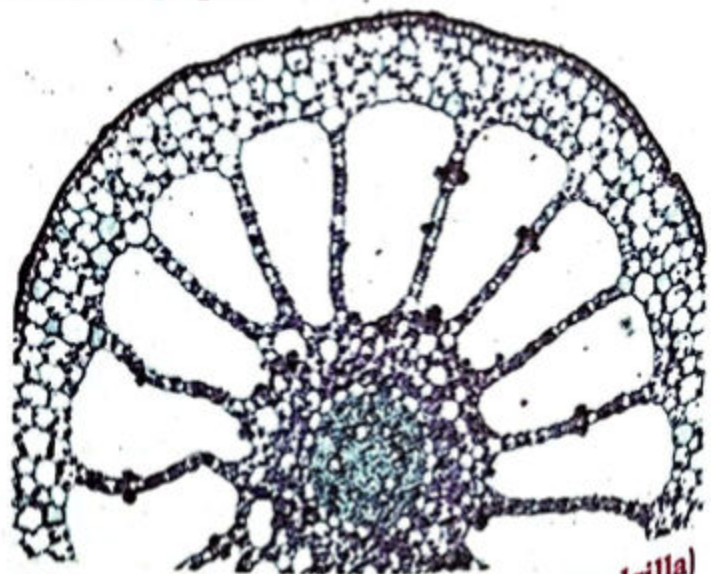


Fig: 10.7. T.S. of Hydrophyte (Hydrilla)



Halophytes:

The plants growing in salt marshes close to sea are termed as **Halophytes**, some of the examples of halophytes are glasswort, cord-grass, etc., they have to absorb water from such a soil which has higher salt concentration therefore they have lower water potential (higher osmotic pressure). Halophytes cope with this situation by roots develop lower water potential which brings in water by osmosis. The excess salt can be stored in cells or excreted out from salt glands on the leaves. The salt thus secreted by some species help them to trap water vapors from the air which is being absorbed in liquid form by leaf cells therefore this is another way for obtaining additional water from the air.

Mesophytes:

These are most of the land plants of temperate zones, which grow in well watered soil they can easily compensate the water lost by transpiration through absorbing water from the soil to prevent excessive transpiration they have developed a water proof external covering called cuticle. In Mesophytes root system is well developed. Stem is erect and branched. Leaves are generally large size. The dorsoventral leaf (dicot) contains stomata in the lower epidermis while upper epidermis has very less number of stomata. Isobilateral leaf (monocot) contains stomata more or less equal in numbers both in upper and lower epidermis.

Xerophytes:

Plants living in dry places such as deserts, steep hills and high altitude have to face scarcity of water they are termed as **xerophytes**. Under such condition water potential xerophytes of soil and air are very low. The xerophytes have developed following adaptations to conserve water and to survive during drought condition.



Fig: 10.8. Xerophyte

Seed/spores as adaptations in life cycle during drought condition:

Land plants produce seeds or spores during their life cycle, the protoplasm's of these structures are quite concentrated and usually protected externally by hard coats and thus these structures remain viable for a considerable period of time.



Adaptations for balance between transpiration and water uptake:

Xerophytes develop very deep vertical roots for the better absorption of water from the soil as seen in acacia, banyan etc.

Other plants such as cacti have superficial horizontal roots which can absorb water before it evaporates from the soil. Reduction in number of stomata, sunken type of stomata development of hairy epidermis, folding of leaves, reduction in size of lamina and modification into spines are some of the adaptations for reducing rate of transpiration.

Some plants store water in larger parenchymatous cells present in stem or leaves. Such plants are termed as succulents. As a result, the stem or leavers become juicy.

10.6.2 Adaptations of plants to low and high temperature:

Low temperature:

Low temperature affects the fluid nature of plasma membrane of plant cell. It is ultimately related to affect the transport of solutes across the membrane. Under such conditions the plants cells increase the proportion of unsaturated fatty acids which prevent crystal formation. The low temperature at the level of freezing point may causes ice crystal formation in cell, this is avoided by the plants inhabiting cold regions by developing freezing tolerance in which the composition of solutes of cell is altered in a way that ice crystal are formed in the cell wall rather than the **cytosol**. The cytosol is super cooled below the freezing point without the formation of ice.

High temperature:

High temperature is more harmful than low temperature since enzymes are denatured which is disastrous for metabolism. The principle way to cool down the plant is transpiration. At 40°C or above most of the plants cells synthesize heat shock proteins that protect enzymes. Plants have some other ways of avoiding overheating such as shiny cuticle, a small leaf surface area, wilting, etc.

10.7 GROWTH AND DEVELOPMENT IN PLANTS:

When an embryo or small seedling comes into being a second phase of the life starts which is called growth. Growth may be defined as a permanent or irreversible increase in size, weight, shape and structure usually accompanied by a permanent change of form. A new born becomes mature when it passed through growth.

In lower plants the entire plant body to specific may be capable of growth. In the higher plants, however growth is confined regions called the growing points. These regions are called meristems. They consist of group of cells, which are capable of divisions and giving rise to new cells. These dividing cells are called meristematic cells. The most important groups of meristematic cells are found at the stem and root apices and constitute the apical meristem.



Role of Apical meristem:

The activity of apical meristem results in primary structure of a plant or the primary growth. They are responsible for the increase in the length of the plant's axis at both the stem and root branches. They are also responsible for the production of lateral appendages such as leaves and floral part. In certain cases, increase in length also takes place by the activity of intercalary meristem. These are part of the apical meristem which have become separated from the apex by permanent tissues and left behind as the apical meristem moves on during growth. The lateral meristems occur at the cambium and phellogen of cork. Activity of these tissues result in the increase in diameter of stem and root the secondary tissues of plants are formed, this increase in thickness due to activity of lateral meristems is called secondary growth.

Turgor pressure:

The whole body of lower plants (Bryophytes) is made up of soft parenchyma tissues. They usually have thin primary walls but no secondary walls. They have a large central vacuole surrounded by a peripheral layer of cytoplasm. They are loosely packed with intercellular spaces in leaves and green herbaceous stem. They contain chloroplasts largely occurs in these cells. They take in water by endosmosis and become extended, this extended parenchyma are turgid, exert an internal pressure called turgor pressure. Due to this turgor pressure these parts remain firm and rigid. If these cells lose water, they also lose turgidity, which causes wilting in herbaceous stem and leaves. Therefore, this turgid parenchyma important for support and shape of the soft plant.

10.7.1 Support in plants:

Young stem have special types of anatomical arrangement, which also helps in supporting plant. The outermost layer of thin walled cells called epidermis. Region beneath epidermis is generally called cortex and the central portion called stele, which is mainly consist of vascular tissues and some soft tissues. This types of stem depends for it's mechanical support in the following tissues:

1. Thin wall parenchyma-having turgidity.
2. Thick walled living tissues like collenchyma and dead tissues like sclerenchyma.
3. Stele the cylindrical core of vascular bundle.

Parenchyma, a kind of simple tissue found in epidermis, cortex and pith. These are relatively unspecialized vegetative cells. The whole body of lower plants (bryophytes) is made up of these tissues. They usually have thin primary walls but no secondary walls. They have a large central vacuole surrounded by a peripheral layer of cytoplasm. They are loosely packed with

intercellular spaces in leaves and green herbaceous stems. They contain chloroplasts therefore photosynthesis largely occurs in these cells. They take in water by endosmosis and become extended, these extended parenchyma are turgid, exert an internal pressure called **turgor pressure**. Due to this pressure these parts remain firm and rigid. If these cells loose water, they also loose turgidity, which causes wilting in herbaceous stems and leaves. Therefore, these turgid parenchyma are important for support and shape of soft plant.

Collenchyma is another type of simple tissue, which is important to provide support in plants. They are also living tissues, more elongated structurally similar to parenchyma except their walls and irregularly thickened. The thickened area is usually more prominent at edges. They function as an important supporting tissue in young plants, in the stem of non-woody older plants and in leaves.

Sclerenchyma another type of supporting tissue. They are simple fundamental dead tissues. They have uniform thickness, heavily lignified secondary walls, which give strength to the plant body. Often these walls are so thick that the lumen of the cells become nearly vanished. Sclerenchyma cells are of two types, fibers and sclereids.

Fibers are elongated cells with tapered ends. They are tough and strong, but flexible while **sclereids** are variable, often irregular in shape. The simple unbranched sclereids are generally called stone cells, they are common in nut and hard parts of the seeds.

Trachieds are elongated tubular heavily lignified dead cells having large hollow cavities. They have oblique transverse walls making trachieds spindle shaped. They are found in xylem acting as supporting tissue in addition to transport of water and dissolved organic salts.

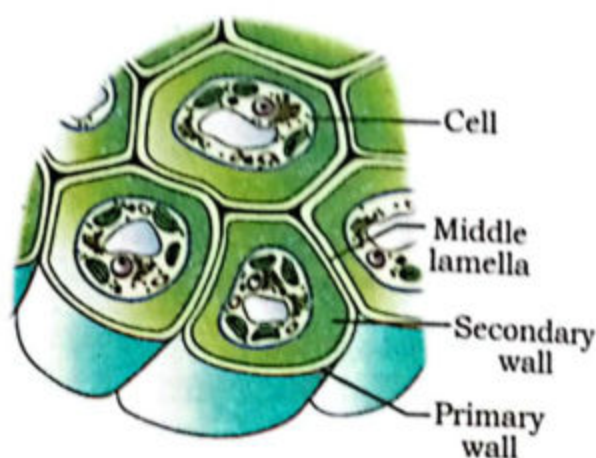


Fig: 10.9. Primary cell wall

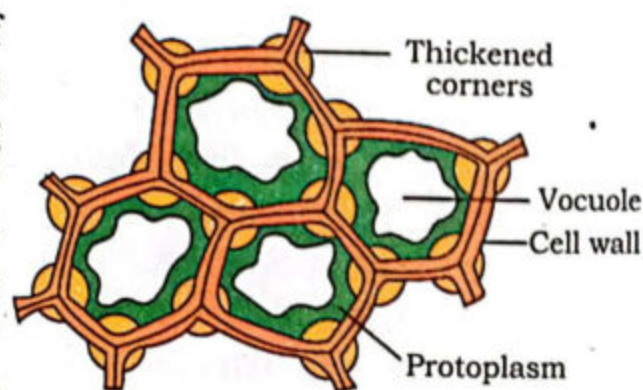


Fig: 10.10. Collenchyma

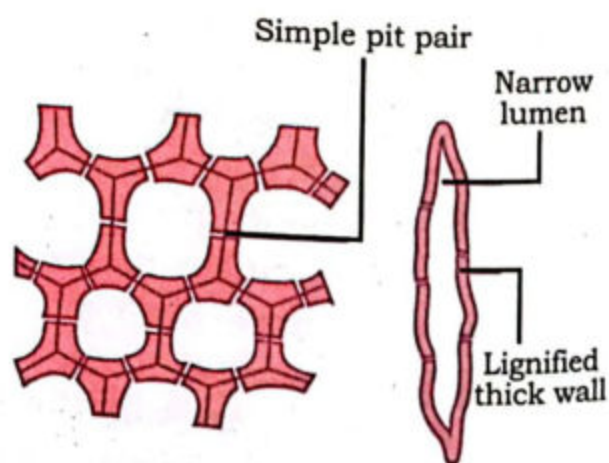


Fig: 10.11. Sclerenchyma



Vessels (trachea) are like trachieds but have no transverse walls and placed end to end forming a structure like open water pipeline. They are also found in xylem acting like trachieds.

Annual rings:

The plants of temperate region accumulate secondary xylem in the form of concentric layers every year and called annual rings.

Each annual ring consists of two zones, the inner zone of **spring wood** having larger vessels and an outer zone of **summer-wood** or **autumn-wood** having smaller vessels.

A fairly accurate estimate of age of old tree can be made by counting annual rings. Study of the rings of large sample of very old tree can also give clue to the climate of an area. Tree ring dating have been used in archeological studies.

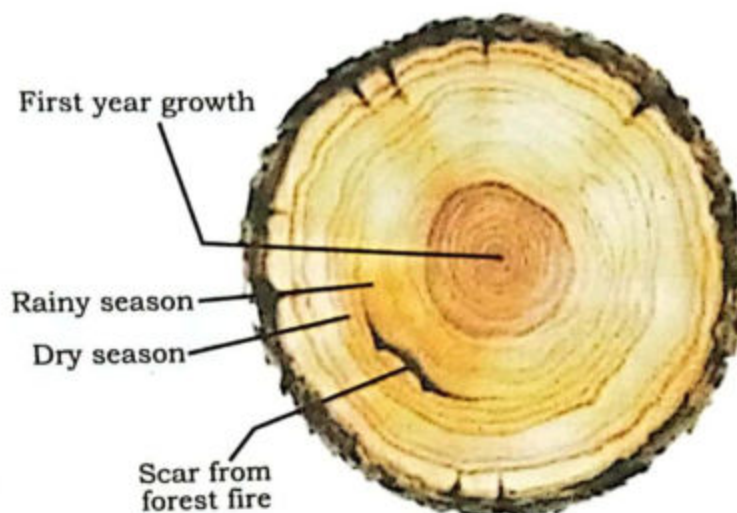


Fig: 10.12. Annual ring

Activity: - Observe annual rings in the log of a tree and calculate age of plants by counting number of annual rings.

10.8 GROWTH RESPONSES IN PLANTS:

10.8.1 Plant hormones/growth regulators:

Certain chemicals produced by plants have a profound effect on their subsequent growth and development. Such chemicals are called plant hormones or phytohormones. They are synthesized by plants in minute concentration and exert their effect either by altering gene expression, activating or inhibiting enzymes or changing properties of the membrane. They are produced in young embryonic tissues as there is no specific organ for their production in plants.

There are five kind of plant hormones. These are:

- i. Auxins
- ii. Gibberellins
- ii. Cytokinins
- iv. Absciscic acid
- v. Ethene

i. **Auxins (Gr. Auxano= To increase):**

Auxins is a class of plant growth substances both natural and synthetic first-revealed by Fritz-Went (1926).

They were the first of the major plant hormones to be discovered and a major coordinating signals in plant development. Indole-acetic acid (IAA) is the principal type of auxin of higher plants, synthesized at the apices of stem and root (apical meristem)

In addition to (IAA) other naturally occurring auxins are: 4-chloro-Indole acetic acid, phenyl acetic acid (PAA) and indole-3 butyric acid (IBA) where the synthetic auxins include naphthalene acetic acid (NAA), 2,4-dichloro- phenoxy acetic acid (2, 4-D) and others.

Auxins co-ordinate development at all levels in plants, from the cellular level to organs and ultimately the whole plant.

Role of Auxins:

(a) **Cell division and cell- enlargement:**

It stimulates cell division, cell enlargement and brings about increase in length of plant. It stimulates wall loosening factor, for example, elastin to loosen the cell-wall. If Gibberellins are also present, the effect is stronger. It also stimulates cell-division if cytokinins are present.

Xylem tissues can be generated when the auxin concentration is equal to the cytokinins.

(b) **Initiation of roots:**

Auxin also initiates development of adventitious roots when applied at the cut base of stem.

(c) **Abscission:**

In mature leaves and fruits when auxin production diminishes, a layer of thin walled cells is formed at the base of petiole and stalk of fruit. This layer is called abscisic layer and causes fall of leaves and fruit with slight jerk.

(d) **Growth of fruit:**

Auxin is responsible for cell division, the tissues of ovary initiate to divide by auxin and the ovary enlarge to develop this process is called fruit setting.

(e) **Parthenocarpy:**

Use of auxin also helps in producing parthenocarpic or seedless fruits.

(f) **Apical dominance:**

Besides growth promoting function, auxin also has inhibitory effect on growth. Growth of apical bud inhibits growth of lateral buds beneath the stem. This phenomenon is termed as apical dominance. Removal of apical bud initiates growth of lateral buds with more leaves and axillary buds.



Weedicide:

- (g) Auxins are selective weed killer. 2-4-dichlorophenoxy acetic acid (2-4-D) is used to kill weeds in lawns and cereal crops.

Flowering:

- (h) Auxin plays a minor role in the initiation of flowering. It can delay the senescence of flowers in low concentrations.

Gibberellins:

- ii. It is a group of chemicals that promote cell division and cell elongation.

First noticed in *Gibberella fujikuroi* fungus which infected rice seedlings and produced a disease called **bakanae** (foolish seedling). The infected seedlings elongated and ultimately fell over without producing grains. Even extract from the fungus when applied to rice seedlings produced the same disease indicating that a definite chemical compound is responsible for disease. **T. Yabuta** and **T. Hayashi** succeeded in isolating the active substance from the fungus and was named Gibberellin, after the name of genus. Its ability to induce growth attracted the scientists who have isolated 70 different types of Gibberellin, many of them occur naturally in higher plants. Major sites of their production are roots, stem and leaves.

Role of Gibberellin:

Gibberellin produces wide variety of effects. One of their effect, like auxin is to stimulate cell division and cell elongation it prevents genetical and physiological dwarfism. They also mobilize food stored in endosperm by producing enzyme (amylase) that converts starch into sugar which is made available to developing embryo.

They also stimulate flowering, fruits development, bud sprouting, growth of pollen tube and parthenocarpy.

iii. Cytokinins:

These are a group of substances both natural and synthetic, which react with auxin to induce cell division. Originally obtained from coconut milk, the other sources are Herring sperm DNA and yeast extract. One of the naturally occurring cytokinin is **zeatin**, which was obtained in pure crystalline form from immature corn grains. **Kinetin**, a synthetic cytokinin has the same effect as that of zeatin.

Role of cytokinin:

They initiate rapid cell division but only in presence of auxin. They also cause delayed senescence (old age). Detached leaves which would normally lose chlorophyll are prevented from becoming decolorized by their application. They also break seed dormancy and promote fruit development in some species.

iv. **Abscissic acid:**

In contrast to growth promoting hormones like auxin, gibberellins and cytokinins, abscissic acid (ABA) is growth inhibitor produced by plant during adverse environmental conditions such as drought condition and at the onset of winter. It induces dormancy in buds and seeds, causes stomata to close, turns leaf primordia into scale which protects the buds and promotes senescence.

v. **Ethene:**

The most important role of ethane (a gas), is that it triggers ripening of fruit. It affects permeability of cell membrane, which allows enzymes responsible for destroying chloroplast with the result that red and yellow colours are unmasked and fruit assumes ripened colour. It contributes to leaf abscission and also breaks dormancy of buds and seeds in some species. It also initiates flowering in some plants e.g. pineapple.

10.9 MOVEMENT IN PLANTS:

All living organisms have the property of movement to fulfill the need of their nutrition, protection, shelter and reproduction. Movement is defined as any action taken by living organism to reduce its irritability produce by stimuli. Plants are usually sessile which respond to environmental factors by adjusting their pattern of growth, development and turgor pressure.

Types of movement in plants:

On the basis of stimuli there are two main types of movement found in plants.

1. Autonomic or Spontaneous movement cause by internal stimuli
2. Paratonic or Induced Movement caused by external stimuli

Here we discussed only Paratonic or induced movement in detail.

10.9.1 Paratonic or induced movement

Movements which occur due to external stimuli are known as induced or paratonic movements. They may be tropic or nastic. The external stimuli which cause these movements may be light, temperature, water, chemicals, gravity etc.

Tropic movement: (Directional movement)

Tropism or tropic movement are growth responses that result in curvatures of whole plant organs towards or away from stimuli.

Movement caused due to external stimuli coming from one side, controlled by the direction of stimulus, respond in the form of growth of curvature in one direction called tropic movement. It is commonly found in radially symmetrical organs of plant such as root and stem. On the basis of stimuli following are the types of tropic movement found in plants.

Phototropism:

It is a curvature movement that takes place when plant is exposed to light coming only from one direction. If this curvature movement is towards



the source of light called positive photo tropism and if away from the source of light called negative phototropism. Phototropism curvature is due to light effect on the distribution of auxin (plant growth promoting hormone)

Geotropism:

It is the movement caused in response to gravitational stimulus. Positive geotropism is observed in the primary roots of many plants and negative geotropism in their shoot.

Chemotropism:

Chemotropism is the movement caused due to chemical for example pollen tube grows through the style towards ovule due to chemical stimulus.

Hydrotropism:

The movement of plant organs in response of water stimulus called hydrotropic movement. This also result in curvature of the organs due to unequal growth on its two sides. Roots are positively hydrotropic.

Thigmotropism:

Thigmotropism is the curvature movement of plant in response to touch stimulus. It can be observed in twiner and climbers. When they touch the solid object the growth on the opposite side of contact increases and they coiled around the support.

Activity:- Demonstrate folding of Mimosa leaf after touch

10.10 PHOTOPERIODISM AND VERNALIZATION:

Photoperiodism and Vernalization serve to synchronize the reproductive behavior of plants with their environment, ensuring reproduction at favorable times of years. They also ensured that members of the same species flower at the same time, encouraging cross pollination for genetic variability.

Photoperiodism:

Length of day light period has marked influence on the behavior of plants particularly on their flowering. The phenomenon in which the influence of photoperiod length on plants is studied is called Photoperiodism. It may also be defined as "response of flowering to relative length of the day and night"

On the basis of their differing responses to light and dark, flowering plants can be divided into three groups,

Long day plants:

Those require long photoperiod and short nights for their flowering. Long day plants flower only when the light period exceeds a certain critical length in each 24 hour cycle. This varies, but an average is about 23 hours to 14 hours. e.g., Hibiscus, Lettuce, Spinach, Radish, Sugar beet and Potato.

Short day plants:

Those that require short days and long nights, for their flowering. Short day plants flower only when the light period is shorter than a critical length in each 24 hours cycle. e.g., Rice, Chrysanthemum, Soya bean, Onion.

Day neutral plants:

Plant which are indifferent to day length for their flowering, for example Tomato, Cotton.

The role of phytochrome in photoperiodism:

The measurement of photoperiod by short and long day plants is now known to depend on a light sensitive pigment, called Phytochrome, which is distributed throughout the plant in minute quantities, being most concentrated in growing tips. Phytochrome exists in two inter convertible forms. The inactive form of Phytochrome P_R (red) is converted to active form P_{FR} , following exposure to red light with peak sensitivity at 660nm. P_{FR} is slowly converted back to the inactive form in darkness or may be rapidly converted to P_R by exposure to far red light with peak sensitivity at 730nm.

Although day light has both red and far-red light but red is predominant, so that in day time, P_R is converted to P_{FR} (far-red) which stores in the leaves. The onset of flowering seems to be determined by the balance between the two forms of Phytochrome pigment, which in some way switches on the genes responsible for flowering.

Vernalization:

Besides having advantageous effects of temperature on the growth of plants, low temperature plays another role i.e., vernalization which may be defined as: promotion of flowering by a cold treatment given to the imbibed seeds or young plants. The effect of low temperature treatment applied in the earlier stages of development has an after effect on the later stages of the development of a plant, many plants require cold treatment for germination and biennial or winter variety. The annual or spring variety is sown in spring and flowers in summers of the same year. The biennial or winter variety is sown in fall and the yield is obtained in the summer of the next year. If the seeds of this variety are sown in spring, the plants do not flower and yield no seeds. It appears that low winter temperatures are necessary for the development of seeds in the winter varieties. Similarly, a biennial like turnip, sugar weed, will not produce flowers if not exposed to cold winter temperatures.



SUMMARY

- Macronutrients are required in large amount, e.g Nitrogen, Phosphorus, Potassium, Sulphur and Sodium etc.
- Micro nutrients, trace elements are required in small amount include manganese, iron, copper, zinc etc
- Carnivorous plant typically live in nitrogen deficient habitats
- Flowering plant exchange gases by diffusion through stomata.
- The intensity of gases exchange depend upon the intensity of respiration
- Some mechanism is needed for transporting substances
- Biologists term the maintenance of steady state as homeostasis
- Movement of water takes place through xylem and mass flow transport of sugar occurs in phloem
- In root transport of water takes place in radial direction it is also termed as lateral transport
- The upward movement of water from roots up to leaves takes place through xylem depend upon two factors transpiration pull and physical property of water i.e. adhesion and cohesion
- Mesophytes are most of the land plant of temperate zone which grow in well watered soil
- The best know plant hormones are auxins, gibberellins, cytokinins, abscisic acid and ethane. they are all effective at very low concentrations
- The measurement of the length of the photoperiod appears to depend on the inter action of phytochrome with and internal clock method.
- Growth may be defined as "permanent irreversible increase in size weight, shape and structure usually anaccompanied by a permanent changes of form.
- The plans of temperate region accumulate secondary xylem in the form of concentric layers every year and called annual ring. A fairly accurate estimate of age old tree can be made by country rings.
- Photoperiodism is the response of plants flowering to relative length of the day and night.
- Vernalization may be defined as promotion of flowering by a cold treatment given to imbibed seeds or young plants
- The translocation of photosynthates always taken place from source to sink tissues so referred as source to sink movement.



EXERCISE

1. Encircle the correct choice

- (i) From where Plants primarily obtain essential nutrients:
(a) Water (b) Air
(c) Soil (d) Light
- (ii) Identify a micro nutrient for plants amongst the following:
(a) Potassium (b) Phosphorus
(c) Iron (d) Sulphur
- (iii) In which mineral deficient soil carnivorous plant typically live:
(a) Nitrogen (b) Calcium
(c) Magnesium (d) Potassium
- (iv) The plants growing in salt marshes close to sea are termed as:
(a) Hydrophytes (b) Xerophytes
(c) Mesophytes (d) Halophytes
- (v) The cause of color of leaf turns in to dull or bluish green is the deficiency of:
(a) Nitrogen (b) Phosphate
(c) Potassium (d) Magnesium
- (vi) Which principle type of auxin of higher plants synthesized at the apices of stem and root:
(a) PAA (b) 24-D
(c) IAA (d) NAA
- (vii) Active transport of sucrose into sieve elements is a step commonly called:
(a) Phloem loading (b) unloading
(c) Diffusion (d) Osmosis
- (viii) The capacity of a living system to lose water is:
(a) Osmotic pressure (b) Water potential
(c) Osmosis (d) Plasma
- (ix) The production of lateral appendages depends on the activity of:
(a) Lateral meristem (b) Intercalary meristem
(c) Apical meristem (d) none of these
- (x) Which process involved in the promotion of flowering by cold treatment:
(a) Photoperiodism (b) Vernalization
(c) Secondary Growth (d) Transpiration



Write short answers of the following questions:

1. Why mineral nutrients are necessary for plants?
2. Why N_2 is included in mineral nutrient although it is not mineral?
3. Why desert plants reduce their leaf size?
4. Why carnivores plants use insect as food?
5. Why phytochrome pigments are important to photoperiodism?
6. How annual rings are formed?
7. Why cold treatment is necessary for the germination of seed in some plants?
8. How osmotic adjustment is beneficial for plants?

Write detailed answers of the following questions:

1. Give an account of exchange of gases in plants?
2. How does pressure flow theory explain the movement of sugar through the plants?
3. Describe the cohesion tension theory of water movement through xylem? What supplies the cohesion and what is the source of tension? How do these two interact to move water through a plant?
4. What essential inorganic nutrient does a plant require for growth, which of these nutrients are required in trace amounts? What are some of the general functions of inorganic nutrients?
5. What are growth regulators? Name and discuss five in detail?
6. Discuss categorization of plants based on osmotic adjustment?